2. Properties of Airborne Contaminants

In the preceding chapter, the characteristics of air were discussed. In this chapter, the discussion will center around the behavior of contaminants that are airborne. Airborne contaminants are those materials that are part of the air mixture but that are foreign to the normal state of the mixture. The contaminants in air may be harmful, irritating, or a nuisance to man. These contaminants may be the result of natural occurrences or the result of man's activity.

The natural contaminants found in the air are legion. For example, ozone, which is formed photochemically or by electrical discharge during storms, can be found in the atmosphere. Hydrogen fluoride, hydrogen chloride, and hydrogen sulfide, resulting from volcanic disturbances, can also be found in the atmosphere. Salt particles from sea water condensation, dust becoming airborne as a result of wind disturbances, bacteria spores, and pollen are also examples of natural contaminants that can be found in the air.

In addition to these natural contaminants, the industrial revolution has brought about the addition of a number of man-made contaminants to the atmosphere. These contaminants include gases (a material which assumes a gaseous state at 25°C and 760 mm Hg), vapors (a material which assumes a liquid state at 25°C and 760 mm Hg), and particulate matter including dusts, fumes, mists, etc., which are dispersed as solid particles in the air (aerosol). It is these man-made contaminants that are of interest to the air pollution engineer and the industrial hygiene engineer, and that will be discussed in this chapter.

Properties of Gases and Vapors

Gases and vapors, like air, obey the laws discussed in the previous chapter (Boyle's Law, Charles' Law, the Perfect Gas Law, Dalton's Law). It is important to realize the significance of Dalton's law as it applies to a mixture of gases and vapors in air. In general, Dalton's Law predicts that a complete mixing action will occur in which a gas or vapor will occupy the same space as the air. A gas that is allowed to escape into the air will tend to diffuse equally throughout the entire area occupied by the air.

One of the properties of a gas or vapor is its density. Density is the mass of a unit volume of a substance expressed as pounds per cubic foot or grams per cubic centimeter. In the preceding chapter, it was pointed out that the density of air is .075 pounds per cubic foot. If one takes the density of air as the base and forms a ratio with the density of any other gas, the specific gravity of that gas is obtained. A similar property, which is

closely related to density, is viscosity. Viscosity describes the tendency of a fluid or gas to resist flow. Gases and vapors generally have very low viscosity and flow easily.

Toxic Properties of Gases or Vapors. Gases and vapors that contaminate the air may be harmless, or they may exhibit toxic properties. If a gas or vapor is known to be toxic, this fact is generally indicated by placing a threshold limit value (TLV) in terms of parts per million of the amount of the gas or vapor that can be present in the air without causing harm to individuals exposed to that contaminant.

It is useful to review the determination of parts per million of a toxic substance present in the air. When the industrial hygiene engineer takes samples of the workroom air, the temperature and pressure are also noted. The measurements are then analyzed to determine the number of milligrams per liter of the substance that are found in the samples. These milligrams per liter can be converted to parts per million by the following formula:

$$(2.2.1) \qquad ppm = \frac{1000 \times 24.45}{\text{molecular weight}} \times \text{concentration (mg/l)}$$

The above formula is based on the fact that 1 gram mole of a perfect gas at 25°C and 760 mm Hg occupies a volume of 24.45 liters. In order to illustrate the above formula and how that formula can be corrected for varying temperature, the following example problems are presented.

Example Problems--Toxicity

Example 1--Suppose analysis of samples of workplace air yielded a concentration of .0074 mg/1 of HCl. Is this a toxic concentration? Assume industrial hygiene standard conditions exist.

Solution

Molecular weight of HCl = 36.47

then

$$ppm = \frac{24.45 \times 10^3}{36.47} \times 0.0074 \text{ mg/1}$$

$$ppm = 670.4 \times .0074$$

$$ppm = 4.96$$

The TLV for HCl is 5 ppm, and the TLV is not exceeded in the above situation.

Example 2--Assume that the above results were obtained at a temperature of 90°F. Is the TLV exceeded?

Solution

$$ppm = \frac{24.45 \times 10^3}{36.47} \times .0074 \times \frac{460 + 90}{460 + 70}$$

$$ppm = 5.15$$

In this case, the TLV of 5 ppm is exceeded. From this example, it is easy to see the effect of temperature on the results obtained in the sample measurements. If correction were not made for the higher temperature, the results would be understated.

Combustibility. Some gases and vapors are combustible. In general the concentration that is required to support combustion is much greater than the TLV level for toxic combustible gases. For this reason, the combustible limits are normally expressed as a percent of the volume of air rather than in parts per million. In terms of combustibility, the values of concern are the flash point, lower explosive limit, and upper explosive limit.

The flash point is the lowest temperature at which a liquid gives off enough vapor to form an ignitable mixture. By this definition, gases do not have a flash point, since they exist as a mixture in the air regardless of temperature. It should be noted that combustion does not occur spontaneously by this definition. An ignition source must be present in order for combustion to occur.

The lower explosive limit (LEL) indicates the minimum concentration of a vapor or gas in the air that will burn if an ignition source is present. The upper explosive limit (UEL) indicates the maximum concentration of the vapor or gas in the air that will burn if an ignition source is available. Concentrations of combustible gases or vapors that exist between these two limits are those that present a potential fire hazard.

Example Problems--Combustibility

Example 1--Hydrogen cyanide is found to be present in the workroom air at a concentration level of 0.015 mg/1. Does this concentration represent a fire hazard?

Solution

Using the previous formula to determine the concentration in parts per million:

$$ppm = \frac{24.45 \times 10^3}{27.03} \times .015 \text{ mg/l}$$

$$ppm = 904.55 \times .015$$

$$ppm = 13.57$$

Percent concentration = .001357

Since the LEL for hydrogen cyanide is 5.6%, the above concentration does not represent a potential fire hazard.

Example 2--A concentration of how many mg/liter of hydrogen cyanide will be necessary to yield a potential fire hazard?

Solution

$$ppm = \frac{24.45 \times 10^3}{27.03} \times mg/l$$

$$mg/l = \frac{5.6 \times 10^4}{904.55}$$

$$mg/l = 61.9$$

If we wish to determine the upper concentration UEL for hydrogen cyanide, the UEL percent is substituted as above giving

$$mg/l = \frac{40 \times 10^4}{904.55}$$

$$mg/l = 442.2$$

Example 3--Does the concentration found in Example 1 represent a toxic hazard? The TLV for hydrogen cyanide is 11 mg/m^3.

Solution

The result obtained in Example 1 yielded 13.6 ppm. Converting the TLV to mg/l

$$mg/l = \frac{11\ mg}{m^3} \times \frac{1\ m^3}{1000\ liters}$$

$$mg/l = 0.011$$

This value can be converted to ppm by

$$ppm = \frac{24.45 \times 10^3}{27.03} \times .011$$

$$ppm = 9.95 \text{ or } 10$$

Thus the TLV for hydrogen cyanide (10 ppm) has been exceeded by the results obtained in Example 1 (13.6 ppm).

The above examples demonstrate the large concentration of a gas that is necessary to be combustible as compared to the TLV if the material is toxic.

For example in an environment where workers are present, the problem that is faced with hydrogen cyanide is not one of a potential fire hazard, but one of a potential toxic hazard.

Motion of Gases and Vapor. When a gas enters the air, a molecular mixture occurs as is predicted by Dalton's Law. One would expect that if the gas is much heavier than air, it would tend to settle out of the air. However, in terms of concentrations of interest to the industrial hygiene engineer, this fact is not true. In general, the mixture acts as if it were air alone. This can be predicted since the effect of gravity is on the average molecular weight of the mixture or the density of a mixture, not on the gas or vapor alone. Consider the following example to determine the specific gravity of a mixture of air and 1000 ppm vinyl chloride.

Molecular weight of vinyl chloride = 62.5

1000 ppm = 0.1%

so that

$.001 \times 62.5 = 0.0625$

$.999 \times 28.94 = 28.91$

molecular wt. of mixture = 28.9725

$$\frac{\text{MW mixture}}{\text{MW air}} = \frac{28.97}{28.94} = 1.001$$

even though

$$\frac{\text{MW vinyl chloride}}{\text{MW air}} = \frac{62.5}{28.94} = 2.16$$

In general, a mixture of a gas or vapor in air at the levels of concentration that are of significance to the industrial hygiene engineer will have a motion similar to the air alone. This can be shown by considering the velocity of a gas falling in air, using the familiar formula for a freely falling body:

$$(2.2.2) \quad v = \sqrt{2g(\sigma_g - \sigma_a)h/\sigma_g}$$

where

g = acceleration due to force of gravity or 32.2 ft/sec^2
σ_g = specific gravity of the mixture
σ_a = specific gravity of the air
h = height of the fall in ft.

Hemeon, W. Plant and Process Ventilation, 2d. ed. New York: Industrial Press, 1963, p. 22.

Using this relationship, the velocity of a mixture of 2000 ppm butane and air can be calculated after falling 1 foot.

Molecular weight of mixture is calculated as:

$$.002 \times 58.12 = .12$$

$$.998 \times 28.94 = 28.88$$

$$29.00$$

$$\sigma_a = \frac{29}{28.94} = 1.002$$

Substituting

$$v = \sqrt{2 \times 32.2(1.002 - 1.0)/1.002} \times 1$$

$$v = .36 \text{ fps or } 21.6 \text{ fpm}$$

Since the minimum air velocities of 15 feet per minute are very likely to be found in almost any industrial area, the results obtained above indicate that any potential settling of the gas will be influenced by the motion of the air. In the above example, we have not considered the further diffusion of the gas within the air which in fact, as a result of Dalton's Law, does take place. Therefore, the velocity shown overstates significantly that which would actually occur.

In summary, then, in the design of an industrial ventilation system to handle gas and vapor contaminants, it is necessary only to control flow of air, since the contaminant gas or vapor will follow the pattern of air flow and thus can be controlled. One caution is necessary when considering a potential accidental contamination of air by large amounts of combustible gases or vapors with relatively high specific gravity. If the contamination results from a spill, it can easily be shown that this gas or vapor will tend to settle out and cause a large concentration to be present at floor level where the potential for an explosion can result. This, however, is a separate problem and should be handled apart from the need to control toxic materials at a level below the TLV.

Properties of Particulate Matter

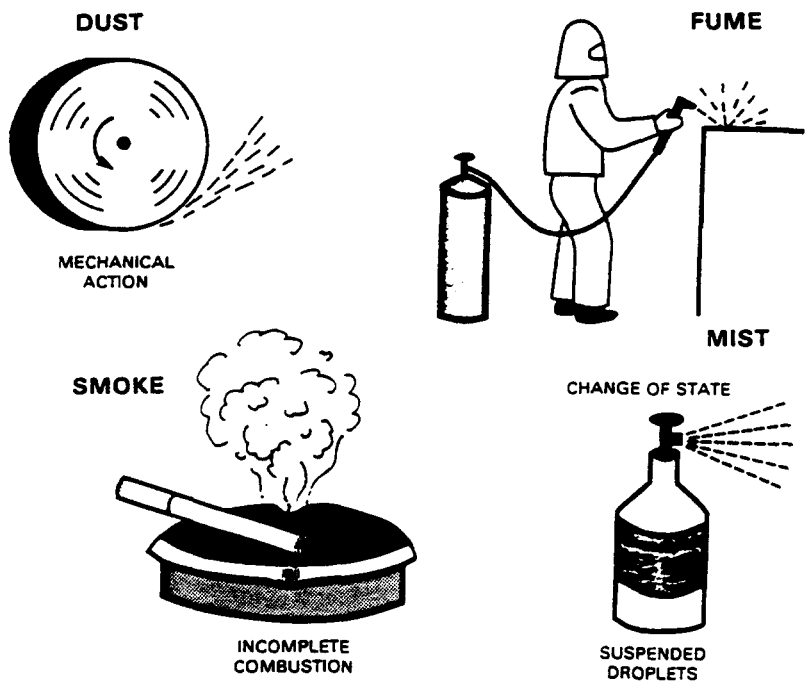
Particulate matter includes dusts, fumes, smoke, and mists that are suspended in the air. This suspension is generally known as an aerosol The presence of particulate matter in the air provides a major problem for the industrial hygiene engineer, particularly if this particulate matter includes toxic materials.

Dust particles that are present in the air are generally formed by a mechanical action. Dusts may be either solid organic or inorganic material. The size of the dust particles varies from the visible to the submicroscopic.

Dust is formed by grinding and crushing operations, movement of crushed materials, and air currents passing over settled dust, causing it to become airborne.

Figure 2.2.1

Particulates in air.



Fumes are formed by a chemical or physical process involving a change of state. Quite often the process requires high temperatures to create the fumes. Typical fumes include oxides of zinc, magnesium, iron, lead, and other metals. Some organic solids may also form fumes by the same processes. Fume particles are very small, normally in the range of 1 micrometer or less.

Smoke refers to airborne particulate matter resulting from incomplete combustion of organic materials. Smoke particles are usually less than one-half micrometer in size when they are produced. Soon after generation, many of the smoke particles will flocculate and form aggregates that appear to be larger particles but are, in fact, many small particles that have grouped together.

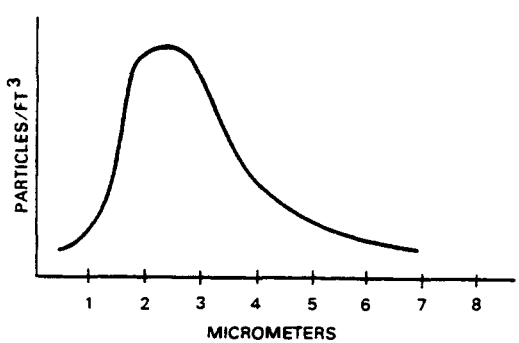
Fibers are a special type of particles in which one dimension is at least three times the length of the other dimension. Fibers can result from either natural fiber-forming materials such as asbestos or cotton or from synthetic materials such as fiberglass. In general, fibers are a special type of dust.

Mists are formed by liquid droplets suspended in the air. Mist may be formed by condensation of a vapor or by spraying which creates an atomized mist. In addition, air bubbles that form on the surface of a boiling liquid can collapse and result in a mist above the liquid. Fog and smog (a combination of smoke and fog) are special cases of a mist.

Particulate Size. Particles that are of interest to the industrial hygiene engineer are limited in size to those particles that are less than 10 micrometers in diameter. This is a result of the fact that particles of a diameter of more than 10 micrometers are not likely to be retained in the respiratory system of the worker. Particles of a size less than 1 micrometer may be retained in the alveoli of the lung. If a sample of air from a particular dust-producing operation is taken, it would be expected that a number of particles of each size would be normally distributed according to size. This is not the case. Since large particles tend to settle out of the air very quickly, the distribution of particles in the air follows what has been shown to be a log-normal distribution with the large number of particles being in the small size range.

Figure 2.2.2

Particle dispersion.



Measurement of particles to determine the number and distribution of size in a given sample is done in a number of ways. The standard method for sizing particles is by counting, using an optical microscope. Sizing is done by measuring using one of three methods:

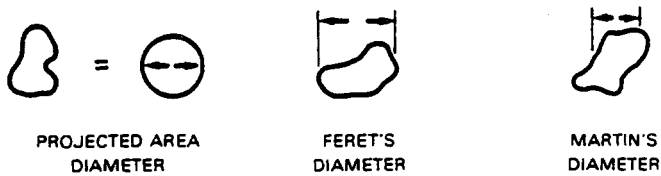
A. Projected Area Diameter--the diameter of a circle of area equal to the profile of the projected area of the particle.

B. Feret's Diameter--the normal distance between two parallel targets to the extreme points of the particle.

C. Martin's Diameter--The distance between opposite sides of a particle, measured in a consistent direction such that the diameter bisects the projected area.

Figure 2.2.3

Particle sizing.



Other methods to measure and count particles are gaining in popularity. Such methods include the electron microscope, impaction with decreasing size orifices, light scattering, and electrostatic charging. However, the standard remains optical sizing because of the low cost and availability of the equipment involved.

Toxicity of Particulate Matter. Particulate matter may exhibit toxic properties. In general, this toxicity is through the respiration of particles. Examples of toxic particulate matter include silica dust, asbestos fibers, carbon tetrachloride mist, mercury mist, beryllium dust, and iron oxide fumes. The threshold limit value for particulate matter is expressed in milligrams per cubic meter. Thus, no conversion need be made to parts per million as in the case of gases and vapors.

Particulate matter has certain optical characteristics such that airborne particles will reflect and refract light. This fact can be easily observed by standing near a sunlit window where the dust particles are obvious as they float through the air. However, particles of the size of interest to the industrial hygiene engineer (10 micrometers or less) cannot be seen by the naked eye. In addition, hazardous concentrations of particulate matter regardless of size are visible in only a few cases and under the best conditions. Thus, observation does not indicate that a safe concentration is present. It is necessary to take samples of the air and analyze them before the safety of the air can be determined.

The Motion of Particulate Matter. The settling rate for dust that is dispersed in the air is determined by the gravitational force acting upon the dust and is a function of the size and specific gravity of the particles. Obviously particles of a larger size and higher specific gravity will tend to settle out of the air more quickly than those with a lower specific gravity.

As the particle settles out of the air, its velocity increases until such a time as it reaches a terminal velocity. As with any other falling body, the frictional force or turbulence that is created by the falling body acts as a frictional drag that tends to retard the acceleration of the body created by the gravitational force. The Reynolds number (R_e) is a dimensionless number that defines the amount of turbulence and the resultant frictional drag that is present. The Reynolds number becomes important in determining not only the

frictional drag and terminal velocity of a falling particle but also the type of flow that will exist within a ventilation duct system. Two types of flow are defined by the Reynolds number. The first of these is streamline flow where there is no turbulence that might retard the flow of the particle or the air in a duct system. The second is turbulent flow where eddies of air are created that act as a frictional drag to retard the flow of the particle or the air. Between streamline flow and turbulent flow a transition flow area is present where the flow is neither entirely streamline nor turbulent.

The Reynolds number can be calculated to obtain a value that indicates in which area of flow the particle is at a given period of time. For a particle Reynolds number of less than or equal to 2, the particle is falling in streamline (laminar) motion. If the Reynolds number is greater than or equal to 2 but less than or equal to 500, the particle is falling in the transition area. Turbulent motion exists where the Reynolds number is calculated to be greater than 500.

The size limit of particles that will be confined to the streamline area can be determined by the following formula:

$$(2.2.3) \quad d_m = \sqrt[3]{103/z}$$

where

d_m = diameter of particle in micrometers
z = the specific gravity of the particle

For a specific gravity of 1, the size limit is 103 micrometers; and for a specific gravity of 3, the size limit is 70 micrometers. Therefore, for particles of industrial hygiene significance, that is, where d_m is less than or equal to 10 micrometers, only streamline motion need be considered.

The terminal velocity of a particle falling in the air in streamline flow can be found by the following equation: (Hemeon, Plant and Process Ventilation, p. 31)

$$(2.2.4) \quad u_t = 9.23 \times 10^{-5} z d_m^2$$

where

u_t = terminal velocity in feet per second
z = specific gravity of the particle
d_m = the diameter of the particle in micrometers

Using the above formula, the terminal velocity of a 10-micrometer particle of specific gravity = 3 when falling through the air can be calculated as follows:

$$u_t = 9.23 \times 10^{-5} z d_m^2$$
$$u_t = 9.23 \times 10^{-5} \times 3 \times 10^2$$
$$u_t = .0277 \text{ ft/sec or } 1.66 \text{ ft/min}$$

It is obvious that the velocity obtained by the particle is very small when compared to random air currents of at least 15 ft/min that occur in an

industrial environment. Given this fact, it is unlikely that particles of the size of interest to the industrial hygiene engineer will settle out of the air. These particles will travel with the air currents and can be controlled by controlling the air itself.

Dust That Is Projected. Many mechanical actions that produce dust also cause the dust to be projected at some speed into the air. For example, a grinding operation will project dust particles in a direction that is tangential to the grinding wheel at the point of contact. Large particles with high velocities will result in large Reynolds numbers indicating high turbulence. As the particle slows down, it will go through a transition state and finally will enter streamline flow. Small particles, because of their small mass, will be projected only a small distance. This projection will occur in streamline flow alone because of the low velocities that are attained. Small particles will be carried in the turbulent motion of the air created by large particles. Small particles that are projected will follow the path of air currents rather than the projected path and, as a result, can be controlled by controlling the air at the source.

Brownian Motion. One final cause of particulate motion in the air, that of Brownian motion, should be briefly discussed. The molecules of air are in constant movement. During this movement, the molecules collide with particles suspended in the air, thus causing a constant random movement of the particle about a point. Small particles are affected greatly by this molecular motion. Brownian motion tends to keep small particles of less than .25 micrometers in diameter in the air. In fact, particles of the .25 micrometer size show a minimum amount of movement because the Brownian motion exactly offsets the gravitational attraction for particles of this size. Therefore, this size particle becomes extremely difficult to remove from the air. However, this fact also makes it difficult for these particles to be retained by the lungs. Particles smaller than 1 micrometer are too small to be affected by molecular action and thus are easily deposited.

Causes of Initial Dispersion of Particulates in the Air

During the previous discussion, the motion of particles after entering the air was investigated. It is of interest to determine what causes the particles to enter the air in the first place. There are a number of causes of particulate dispersion in the air, some of which have been alluded to in the earlier discussion.

The first cause of particulate dispersion is a mechanical cause resulting from grinding or cutting of a material. The grinding wheel or saw, as it cuts through the material, projects particles in a line tangent to the point of contact between the material and the mechanical wheel or saw. The smaller particles are picked up by air currents and carried through the air to other locations. Other examples of this type of mechanical action include drill presses, lathes, and sanders.

A second cause of dispersion is shattering. Consider hitting a rock with a hammer. Small chips and particles are broken away from the rock; and, if

these particles are small enough, they will remain in the air and follow existing air currents.

Falling particles create a splash when they impinge upon a surface. This splash can be likened to an explosion where the particles tend to be pushed outward and upward away from the point of contact. At this point, passing air currents can hold the smaller particles in suspension.

Prior to impinging upon a surface, a stream of falling particles will create turbulence. Air currents resulting from the turbulence will carry small particles outward from the falling particulate stream. These small particles can then be dispersed in the air.

Thermal convection provides another method wherein particulate matter can become dispersed in the air. As hot air rises, it can carry with it fumes, mists, or particulate matter. These contaminants can then become further dispersed in the air as cooling and mixing in the atmosphere occurs.

The final method by which particulate matter can become dispersed in the air is by a secondary air movement. Consider a passing automobile. Turbulence created by the automobile causes swirls of dust and loose papers to become airborne. The papers will settle out quickly; however, the small dust particles may remain airborne for long periods of time as a result of this turbulent action. Certain types of machines produce a fan action that results in air currents about the machine. These air currents can cause particulates to be carried away from the source where the smaller particles will be subjected to further dispersion by the random air currents in the work environment.

Summary

From the preceding discussion, it can be seen that gases, vapors, and particulate matter of interest to the industrial hygiene engineer can be controlled by controlling the air in which they are suspended. This is an important fact to remember when designing an industrial ventilation system. Certain special operations require that larger particles be captured that have projected motion or that tend to settle quickly. The design of such a system involves different considerations from those that are necessary to control toxic contaminants in the air. The fact that the particles, gases, and vapors present in the air can be controlled by controlling the air tends to simplify the design of industrial ventilation systems. However, quite often this characteristic is not taken advantage of in the system design. As a result, the system may be over-designed for the job that is required.